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Paper 39
Date: March 23, 2026

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

BIO-RAD LABORATORIES, INC.,
Petitioner,

v.

CALIFORNIA INSTITUTE OF TECHNOLOGY,
Patent Owner.

IPR2024-01451
Patent 11,827,921 B2

Before GRACE KARAFFA OBERMANN, TAWEN CHANG, and CYNTHIA M. HARDMAN, *Administrative Patent Judges.*

CHANG, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining No Challenged Claims Unpatentable
35 U.S.C. § 318(a)
Denying Patent Owner's Motion to Strike in Part and
Dismissing it in Part as Moot
37 C.F.R. § 42.64
Dismissing Petitioner's Objections to Demonstratives as Moot
37 C.F.R. §§ 42.64, 42.70

I. INTRODUCTION

This is a Final Written Decision in an *inter partes* review challenging the patentability of claims 1–19 of U.S. Patent No. 11,827,921 B2 (Ex. 1001, "the '921 patent"). We have jurisdiction under 35 U.S.C. § 6.

Petitioner has the burden of proving unpatentability of the challenged claims by a preponderance of the evidence. 35 U.S.C. § 316(e). Having reviewed the parties' arguments and cited evidence, for the reasons discussed below, we find Petitioner has not demonstrated by a preponderance of the evidence that claims 1–19 are unpatentable.

A. Procedural History

Petitioner Bio-Rad Laboratories, Inc. filed a Petition requesting *inter partes* review of claims 1–19 of the '921 patent. Paper 1 ("Pet."). Patent Owner, California Institute of Technology, timely filed a Preliminary Response. Paper 8 ("Prelim. Resp."). The parties further submitted an authorized Reply and Sur-Reply to the Preliminary Response. Paper 9; Paper 10; Ex. 3001. In view of the then-available preliminary record, we instituted an *inter partes* review. Paper 11 ("Inst. Dec.").

After institution, Patent Owner filed a Response to the Petition. Paper 17 ("PO Resp."). Petitioner filed a Reply. Paper 20 ("Reply"). Patent Owner filed a Sur-reply. Paper 27 ("Sur-reply"). At the Board's request, the parties further filed supplemental briefs on the construction of the term "single sample volume." Ex. 3003; Papers 25, 29, 30, 33.

On October 24, 2025, Patent Owner filed a Motion to Strike Portions of Petitioner's Reply Pursuant to 37 C.F.R. § 42.23(b). Paper 26 ("Mot."). Petitioner filed its Opposition to the Motion on October 31, 2025. Paper 28 ("Opp.").

On December 31, 2025, the parties filed their respective demonstratives. Papers 35, 36. Petitioner filed Objections to Demonstratives on January 2, 2026. Paper 37 ("Obj."). On January 6, 2026, we held an oral hearing, the transcript of which is of record. Paper 38 ("Tr.").

B. Real Parties-in-Interest

Petitioner identifies itself as the real party-in-interest. Pet. 19. Patent Owner identifies itself as the real party-in-interest. Paper 7, 2.[1]

C. Related Matters

Petitioner and Patent Owner indicate the ’921 patent is at issue in the pending district litigation proceeding *In Re: ChromaCode Litigation*, Case No. 5:23-cv-04823-EKL (N.D. Cal.) ("ChromaCode Litigation"), filed on September 20, 2023. *See, e.g.*, Pet. 13, 2; Paper 15, 1. The ChromaCode Litigation also involves U.S Patent Nos. 10,068,051 ("the ’051 patent"), 10,770,170 ("the ’170 patent"), and U.S. Patent No. 12,168,797 ("the ’797 patent").[2] Paper 15, 1.

The Board denied institution of petitions for *inter partes* review challenging the ’051 patent and the ’170 patent, IPR2024-01177 and IPR2024-01178 respectively, on January 29, 2025. IPR2024-01177,

[1] Patent Owner originally identified itself and ChromaCode Inc. as the real parties-in-interest. Paper 4, 2. However, on December 30, 2024, Patent Owner filed Updated Mandatory Notices indicating that ChromaCode Inc. "is now a non-exclusive licensee of Patent Owner and has not controlled or funded this Inter Partes Review proceeding." Paper 7, 2.

[2] The ’797 patent was at issue in a separate district court proceeding, which has been consolidated with the ChromaCode Litigation. Paper 15, 1; *California Institute of Technology v. Bio-Rad Labs., Inc.*, Case No. 5:25-cv-01701-EKL (N.D. Cal.), ECF No. 27.

Paper 10; IPR2024-01178, Paper 10. Petitioner's Request for Rehearing of the Institution Decision in IPR2024-01178 was denied on May 22, 2025. IPR2024-01178, Paper 13.

A petition for *inter partes* review challenging the '797 patent was filed on September 15, 2025; the Director denied the petition on March 10, 2026.[3] IPR2025-01546, Paper 14, 2.

D. The '921 Patent and Relevant Background

The '921 patent is titled "Signal Encoding and Decoding in Multiplexed Biochemical Assays." Ex. 1001, code (54). The '921 patent was filed on May 1, 2020, and claims priority to application No. 15/914,356 (now U.S. Patent No. 10,770,170), filed on March 7, 2018; application No. 14/451,876 (now U.S. Patent No. 10,068,051), filed on August 5, 2014; application No. 13/756,760 (now U.S. Patent No. 8,838,394), and provisional application No. 61/594,480, filed on February 3, 2012. Ex. 1001, codes (22), (60), (63).[4]

In one aspect, the '921 patent relates to methods and kits for the multiplexed detection of analytes in a sample. Ex. 1001, 1:51–52, 8:63–64. The '921 patent states that "[f]luorescence detection has been a preferred technique for multiplexed assays"; however, because "standard fluorophores have wide emission spectra," "only a relatively small number

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[3] Neither Petitioner nor Patent Owner updated their mandatory notices to include IPR2025-01546. The parties are reminded of their obligation under 37 C.F.R. § 42.8(a)(3) to timely update these notices.

[4] Petitioner and Patent Owner both assume a February 3, 2012 priority date. Pet. 6 (assuming, "[f]or the purposes of only this proceeding, . . . that the '921 patent is entitled to a February 3, 2012, priority date"); Ex. 2018 ¶ 14 (Dr. Struhl stating that he understood "the time of the invention" to be "on or before February 3, 2012"). For purposes of this Decision, we apply a February 3, 2012 priority date.

of colors (e.g., 4 to 6) are typically used simultaneously in multiplexed fluorescent assays" in order to "preserve spectral resolution." Ex. 1001, 8:33–34, 8:40–44.

According to the '921 patent,

> [t]he traditional encoding method for multiplexed fluorescent assays has been to encode each analyte with a single color; i.e., M=N, where M is the number of analytes that can be detected and N is the number of spectrally resolved fluorescent probes. Whenever higher factors of multiplexing are required (i.e., M>N), fluorescence is generally combined with other techniques, such as aliquoting, spatial arraying, or sequential processing. These additional processing steps are labor-intensive and frequently require relatively expensive and complex optical and mechanical systems . . . . Such systems are often impractical to deploy in certain settings . . . . Thus, there is a significant need for multiplexed encoding and decoding methods that can provide an inexpensive means of multiplexing while avoiding the use of expensive additional processing steps.

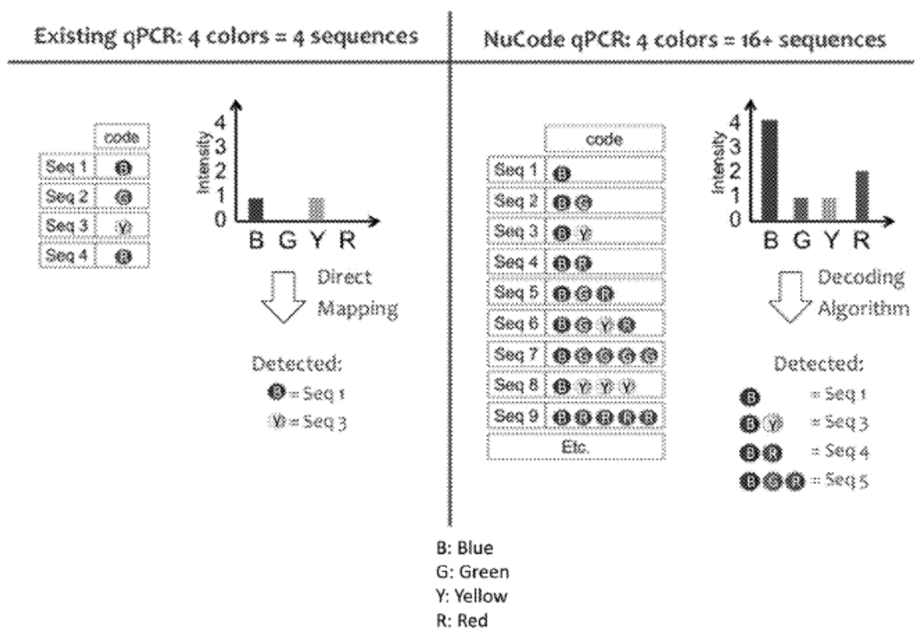
Ex. 1001, 8:45–62.

The '921 patent purports to describe "encoding, analysis, and decoding methods" that permit the detection of a number of analytes greater than the number of spectrally resolved fluorescent probes. *See, e.g.*, Ex. 1001, 8:65–66, 12:39–47.

Figure 1, reproduced below, "shows a comparison between a traditional encoding method of detecting four analytes with four colors" — "B, G, Y, and R, which indicate blue, green, yellow, and red, respectively"

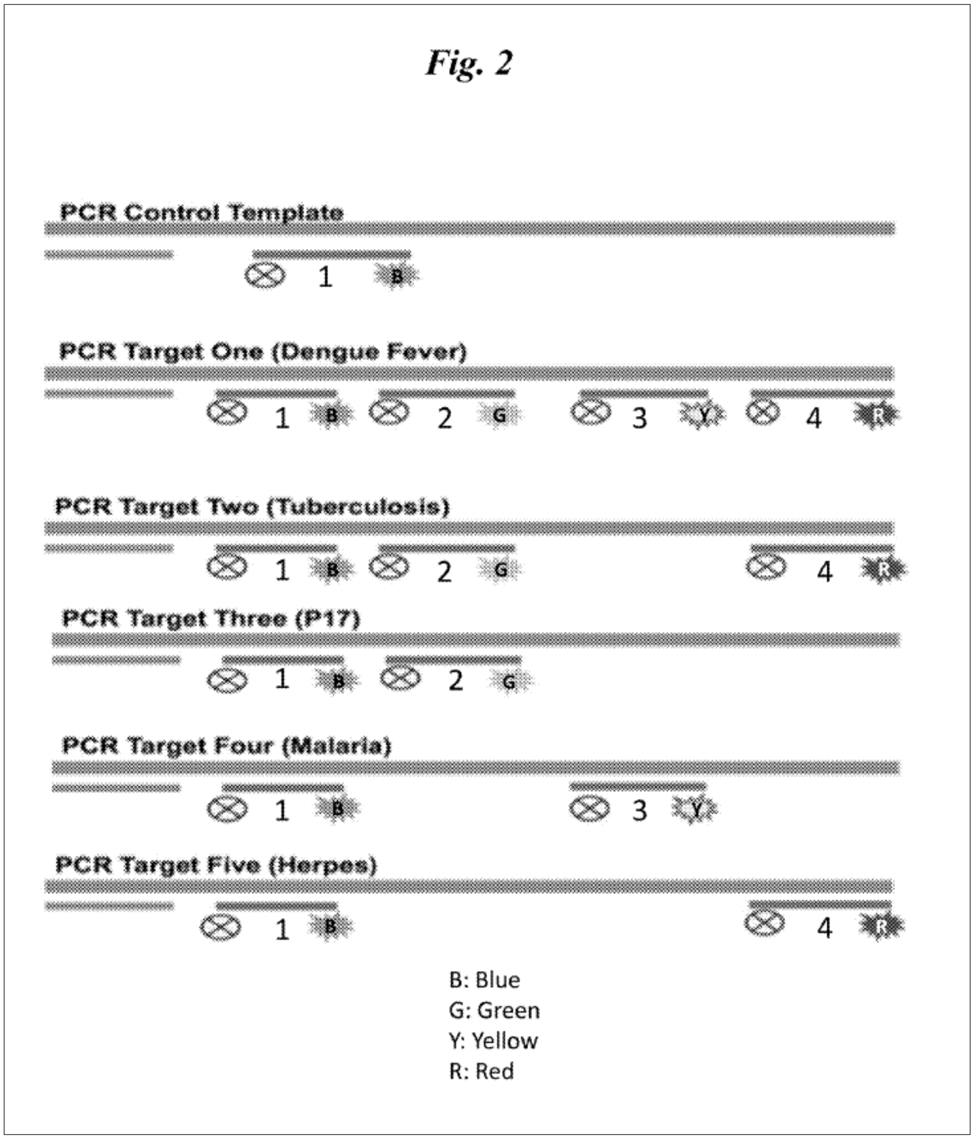
— and “an encoding method of the invention able to detect **16** or more sequences with four colors”:

Fig. 1



Id. at Fig. 1, 7:62–65.

Figure 2, reproduced below, shows "a schematic representation of detection of six analytes (including control) with four colors":



Ex. 1001, Fig. 2, 8:1–2.

Figure 2 illustrates the detection of analytes "using hybridization probes attached to a fluorophore (B, G, Y, R; blue, green, yellow, and red, respectively) and a quencher (oval with 'X')."[5] Ex. 1001, 8:2–5. The '921

[5] As the '921 patent explains, "oligonucleotide probe attached to a fluorophore and a quencher" is sometimes also referred to as a "TAQMAN

patent explains that, in Figure 2, "[t]he control sequence and the five other sequences" specific for each analyte "are all detected using a probe labeled with a blue fluorophore," and "[t]he non-control analytes are each detected using 1-3 additional probes." *Id.* at 8:5–11.

The '921 patent further explains that in some embodiments, "each analyte to be detected is encoded as a value in each of at least two components of a signal," e.g., intensity in addition to wavelength (i.e., color). Ex. 1001, 9:3–5. These components may be "cumulatively measured for the sample" and the "presence or absence of an analyte may then be determined based on the values of each of the at least two components of the signal and the values used to encode the presence of the analyte (i.e., those values in the coding scheme)." *Id.* at 9:39–47.

The '921 patent explains that the terms "degenerate" and "degeneracy," as used in the Specification, "generally describe a situation where a legitimate result is not definitive, because it can indicate more than one possibility in terms of the presence or absence of an analyte." Ex. 1001, 16:47–50. According to the '921 patent, its disclosure further provides methods to reduce or eliminate degeneracy from the coding

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probe." Ex. 1001, 10:31–32. These probes are able to detect the presence of an analyte in a sample because, "[s]o long as the quencher and the fluorophore are in proximity, the quencher quenches the fluorescence emitted by the fluorophore upon excitation by a light source." Ex. 1001, 10:36–39. However, when the probe hybridizes to an analyte in a nucleic acid amplification reaction comprising primers (e.g., a polymerase chain reaction), the extension of the primers by a DNA polymerase degrades the probe and breaks the proximity between the fluorophore and the quencher so that the signal from the fluorophore is no longer quenched. *Id.* at 10:42–50. Accordingly, "the amount of fluorescence detected is a function of the amount of analyte present." *Id.* at 10:50–52.

scheme, "thereby increasing the confidence with which an analyte is detected." *Id.* at 16:57–59. The '921 patent describes different solutions for degeneracy, such as encoding individual analytes using intensity levels so that every possible combination of analytes has a unique cumulative intensity. *See id.* at 23:38–56.

E. Overview of Challenged Claims

Petitioner challenges claims 1–19 of the '921 patent. Claims 1 and 14, reproduced below, are the only independent claims:

> 1. An assay that is capable of unambiguously detecting the presence or absence of each of at least M analytes,
>
> wherein each of said at least M analytes is encoded as a value of one component of a fluorescent signal, thereby generating a coding scheme,
>
> wherein said fluorescent signal comprises F wavelength components, and
>
> wherein said assay is capable of unambiguously detecting the presence or absence of each of said at least M analytes, in any combination of presence or absence, in a single sample volume, based on detection of said F wavelength components, when M is greater than F,
>
> wherein the assay does not require the use of mass spectrometry or immobilization of said at least M analytes. and
>
> wherein the analytes are amplified in the single sample volume.
>
> 14. A kit for detecting the presence or absence of at least M analytes by generating a single cumulative measurement, comprising analyte-specific reagents, wherein:
>
> a. each of said at least M analytes is encoded as a value of one component of a fluorescent signal, thereby generating a coding scheme, wherein each of said at least M analytes is represented in said

> coding scheme by said value, and wherein said encoding is performed in a manner that eliminates degeneracy; and
>
> b. each of said analyte-specific reagents, when contacted with a sample comprising its corresponding analyte, generates the value;
>
> wherein said kit is capable of unambiguously detecting the presence or absence of each analyte of M analytes in a single sample volume, in any combination of presence or absence, and
>
> wherein the kit does not require the use of mass spectrometry or immobilization of said M analytes, and
>
> wherein the analytes are [a]mplified[6] in the single sample volume.

Ex. 1001, 67:10–26, 68:19–38.

Claims 2–13 depend directly or indirect from claim 1 and recite additional limitations relating to the coding scheme, analytes, reagents, fluorescent signal, and/or use of a computer or computer network in the assay. Ex. 1001, 67:26–68:18.

Claim 15 depend from claim 14 and further requires that the "fluorescent signal comprises F wavelength components" and that the kit be "capable of unambiguously detecting the presence or absence of M analytes, in any combination of presence or absence, when M is greater than F." Ex. 1001, 68:39–43.

Claims 16–19 depend directly or indirectly from claim 14 and recite additional limitations relating to the analyte-specific reagents in the kit. Ex. 1001, 68:44–53.

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[6] As both parties recognize (*see, e.g.*, Pet. 78; Ex. 2018 ¶ 46), claim 14 as set forth in Exhibit 1001 appears to contain a typographical error, i.e., "wherein the analytes are *simplified* in the single sample volume." Ex. 1001, 68:37–38 (emphasis added).

F. Asserted Grounds of Unpatentability

Petitioner asserts the following grounds of unpatentability (Pet. 21):

Ground	Claims Challenged	35 U.S.C §[7]	Reference(s)/Basis
1	1–10, 13–19	102	Larson[8]
2	4, 6, 7, 14–19	103(a)	Larson
3a[9]	1–10, 12–19	103(a)	Larson
3b	1–10, 12–19	103(a)	Larson, Saxonov[10]
4	11	103(a)	Larson, Saxonov, Silverbrook[11]

Petitioner relies on supporting evidence including the declarations of Carl A. Batt, Ph.D. (Ex. 1002, "Batt Declaration" and Ex. 1043, "Batt Reply Declaration"). Patent Owner relies on supporting evidence including the

[7] The Leahy-Smith America Invents Act ("AIA"), Pub. L. No. 112-29, 125 Stat. 284, 287–88 (2011), amended 35 U.S.C. § 103, effective March 16, 2013. Because the '921 patent has an effective filing date before March 16, 2013, the pre-AIA version of § 103 applies. Our decision, however, would be the same under either version.

[8] Larson et al., US 2011/0250597 A1, published Oct. 13, 2011 (Ex. 1017, "Larson").

[9] Ground 3 as recited in the Petition is styled as "Obviousness in View of Larson and Saxonov." Pet. 21, 39. However, our review of the Petition shows that Petitioner in fact argues that claims 1–10 and 12–19 are obvious over Larson alone *or* in combination with Saxonov. *See, e.g.*, Pet. 51. For clarity, we designate Petitioner's asserted ground of invalidity over Larson alone as Ground 3a, and the asserted ground of invalidity over Larson and Saxonov as Ground 3b.

[10] Saxonov et al., US 2013/0040841 A1, published Feb. 14, 2013 (Ex. 1004, "Saxonov").

[11] Silverbrook et al., US 2011/0312704 A1, published Dec. 22, 2011 (Ex. 1009, "Silverbrook").

declaration of Kevin Struhl, Ph.D. (Ex. 2018, "Supplemental Struhl Declaration").[12]

II. ANALYSIS

A. Principles of Law

"In an [*inter partes* review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable." *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3) (2012) (requiring *inter partes* review petitions to identify "with particularity . . . the evidence that supports the grounds for the challenge to each claim")). This burden of persuasion never shifts to Patent Owner. *See Dynamic Drinkware, LLC v. Nat'l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015) (discussing the burden of proof in *inter partes* review). To prevail, Petitioner must demonstrate unpatentability by a preponderance of the evidence. 35 U.S.C. § 316(e).

A claim is unpatentable under 35 U.S.C. § 103(a) if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. *See KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) objective evidence of

[12] Patent Owner also relied on a declaration of Dr. Struhl (Ex. 2001, "Struhl Declaration") in support of Patent Owner's Preliminary Response. However, Patent Owner cites only to the Supplemental Struhl Declaration, and not the Struhl Declaration, in the Patent Owner's Response.

nonobviousness, if any. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

An obviousness determination based on a combination of teachings requires finding "a motivation to combine accompanied by a reasonable expectation of achieving what is claimed in the patent-at-issue." *See Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1367 (Fed. Cir. 2016). "[A]ny need or problem known in the field of endeavor at the time of invention and addressed by the patent can provide a reason for combining the elements in the manner claimed." *KSR*, 550 U.S. at 419–20.

B. Level of Ordinary Skill in the Art

We consider the grounds of unpatentability in view of the understanding of a person of ordinary skill in the art (sometimes abbreviated herein as "POSA") as of February 3, 2012. *See supra* p. 4, n.4.

Relying on the Batt Declaration, Petitioner asserts that a POSA as of February 3, 2012 "would have had a Ph.D. in molecular biology, genetics, biochemistry, or a related discipline, with at least a year of experience in quantitative and/or dPCR and fluorescence-based analyte detection," or, alternatively, would have been "someone with greater relevant experience sufficient to compensate for having less relevant formal education." Pet. 9 (citing Ex. 1002 ¶¶ 66–68). Petitioner asserts that

> [a] POSA would have understood the following technologies and techniques:
>
> - Multiplex PCR and its application in detecting and measuring polynucleotide analytes.
> - The design and use of fluorescent-labeled oligonucleotide probes, including use of quenchers, and use in conjunction with multiplex PCR.
> - The quantitative detection and measurement of polynucleotide analytes using fluorescent-labeled

oligonucleotide probes in multiplex applications.
- Techniques for labeling probes and detecting multiple targets using fewer differently colored fluorophores than polynucleotide targets.
- Quantitative PCR and digital PCR and their application in detecting and measuring polynucleotide analytes, including in multiplex applications.

Pet. 9–10 (citing Ex. 1002 ¶ 69).

Patent Owner states that, "[f]or purposes of this IPR proceeding, PO does not contest Petitioner's description of qualifications for a person of ordinary skill in the art (POSA) set forth [above]." PO Resp. 23.

In view of the above, and because Petitioner's uncontested description of one of ordinary skill in the art is reasonable and consistent with the '921 patent and the cited art, we adopt Petitioner's definition of a POSA for purposes of this Decision, except that we replace the phrase "*at least* a year of experience in quantitative and/or dPCR and fluorescence-based analyte detection" with "*around* a year of experience in quantitative and/or dPCR and fluorescence-based analyte detection." Pet. 9 (emphasis added).[13]

C. Claim Construction

In the Petition and Preliminary Response, Petitioner and Patent Owner proposed explicit constructions for the terms "non-degenerately," "degenerate," and "degeneracy." Pet. 22; Paper 8, 27. Because the constructions of these terms are not in controversy — the parties' proposals were substantially similar — and are not needed to resolve the issues in this case, we need not, and do not, construe these terms for the purposes of this

[13] As noted in our Institution Decision, the open-ended language "at least" expands the range of experience indefinitely with no upper bound. Inst. Dec. 14. We therefore do not apply that aspect of Petitioner's definition.

Decision.[14] *See Realtime Data, LLC v. Iancu*, 912 F.3d 1368, 1375 (Fed. Cir. 2019) ("The Board is required to construe 'only those terms . . . that are in controversy, and only to the extent necessary to resolve the controversy.'") (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999)).

With respect to remaining claim terms, Petitioner asserts that, for purposes of this proceeding, the Board should understand the claims "consistent with the apparent interpretations that Patent Owner is pursuing in parallel district court litigation, particularly in view of its contentions regarding how the '921 Patent is allegedly infringed." Pet. 22 (citing Ex. 1012). Patent Owner asserted in its Preliminary Response that these terms should instead be construed in view of the intrinsic record. Paper 8, 28–29. Because the parties did not raise any specific claim terms as requiring explicit construction in the Petition and the Preliminary Response, and because we determined that it was unnecessary to more explicitly construe any claim term to decide whether to institute an *inter partes* review,

[14] Claim construction orders were issued in ChromaCode Litigation on July 22, 2025 and January 30, 2026. Ex. 2024; IPR2025-01546, Ex. 1038. The only terms construed in the '921 patent relate to degeneracy, and the Court's constructions of these terms were substantially similar to those proposed by the parties. Ex. 2024, 26. For the sake of completeness we note that "[t]he Court in ChromaCode Litigation also construed the term "F" in the '797 patent, which like the '921 patent claims priority from application No. 15/914,356 (now U.S. Patent No. 10,770,170), as having its "[p]lain and ordinary meaning." IPR2025-01546, Ex. 1038, 12. However, "F" is used differently in the claims of the '921 and '797 patents. *Compare* Ex. 1001, 67:14–15 ("wherein said fluorescent signal comprises F wavelength components," and the number of analytes M is greater than F), *with* IPR2025-01546, Ex. 1001, 78:39 (where the number of analytes M is equal to C*log2 (F+1)).

we instituted the *inter partes* review without explicitly construing any terms. Inst. Dec. 15–16. *Realtime Data*, 912 F.3d at 1375.

The parties have now placed the limitations "wherein analytes are amplified in the single volume" and "F wavelength components" into controversy. *See, e.g.*, Papers 25, 29, 30, 33; Reply 7–9; Sur-reply 11–13. Accordingly, we construe these terms below.

Because we construe claims in AIA proceedings "using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. [§] 282(b)," 37 C.F.R. § 42.100(b), we construe these terms in accordance with their ordinary and customary meaning, as would be understood by a person of ordinary skill in the art, at the time of the invention, in light of the language of the claims, the specification, and the prosecution history of record. *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–19 (Fed. Cir. 2005) (en banc).

1. "wherein analytes are amplified in the single sample volume"

The parties do not dispute that "single sample volume" encompasses both "a pre-partitioned sample (*e.g.*, within a tube or bottle) or post-partitioned sample (*i.e.*, a droplet)." Paper 29, 1. Neither do the parties appear to dispute that claims 1 and 14 respectively require an assay or kit capable of unambiguously detecting the presence or absence of at least M analytes in the single sample volume in which analytes are amplified. *See, e.g.*, *id.* at 2 ("Regardless of whether the 'single sample volume' is a droplet or a tube or bottle, the 'wherein' clause requires that '***the*** single sample volume' must be where amplification occurs."); Reply 10 ("The claim language requires detecting 'M analytes' in a 'single sample volume,' with amplification occurring in that same volume.").

However, Patent Owner asserts that, once the initial sample is partitioned into droplets, the droplets—whether individually or as a collection—are no longer the same "single sample volume" as the initial sample. *See, e.g.*, Paper 29, 1–2. In contrast, Petitioner takes the position that, after a sample volume is separated into droplets, the collection of droplets "constitute the same 'single sample volume'" as the initial, pre-partitioned sample. Reply 23 ("After the sample volume is separated into droplets, Larson clearly states that 'the droplets are collected off chip . . . in a PCR thermal cycling tube'—which constitutes the same 'single sample volume'—'and then thermally cycled in a conventional thermal cycler.'").[15]

In short, the claim construction question at issue is whether the collection of droplets of a partitioned sample is the same "single sample volume" as the earlier, pre-partitioned sample. We agree with Patent Owner that it is not.

We begin with "the context in which a term is used in the asserted claim," which can be "highly instructive." *Phillips*, 415 F.3d at 1314.

In this case, claim 1 recites that the claimed assay is "capable of unambiguously detecting the presence or absence of each of said at least M analytes, in any combination of presence or absence, in *a* single sample volume, . . . wherein the analytes are amplified in *the* single sample

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[15] In this regard, we note that Petitioner's claim construction position is not predicated on the collection of droplets being reconstituted into its original, pre-partition form prior to amplification; rather, the droplets remain compartmentalized from each other in the PCR thermal cycling tube. Tr. 65:21–66:9 (Petitioner's counsel explaining that the droplets do not combine in the PCR thermal cycling tube but, rather, are "suspended in an immiscible fluid and analyzed en masse").

volume." Ex. 1001, 67:8–25 (emphasis added). The use of the article "the" indicates that the first instance of single sample volume, i.e., the single sample volume in which the presence or absence of each M analytes are to be detected, is the antecedent basis for the second instance of "single sample volume," i.e., the single sample volume in which the analytes are amplified. Thus, as the parties appear to agree, the single sample volume in which ***amplification*** takes places must be the same single sample volume in which the presence or absence of analytes are to be detected. We are not persuaded that, when accorded its plain and ordinary meaning, a single sample volume divided into a collection of droplets would be considered the "same" single sample volume prior to its partition.

We next turn to the Specification, which "is always highly relevant to the claim construction analysis." *Phillips*, 415 F.3d at 1315 (quoting *Vitronics Corp. v. Conceptronic, Inc.*, 90 F.3d 1576, 1582 (Fed. Cir. 1996)). The Specification distinguishes the claimed invention from traditional encoding methods that require fluorescence to be combined with other techniques such as aliquoting, which implicate "relatively expensive and complex optical and mechanical systems" (e.g., microfluidics and drop generators) when the number of analytes is greater than the number of spectrally resolved fluorescent probes. Ex. 1001, 8:45–62. For instance, the Specification notes that, "[i]mportantly, the methods and compositions provided herein may be used to detect multiple analytes by obtaining a cumulative measurement on a single solution" and that "[n]o separation is necessary." *Id.* at 35:66–36:3; *see also id.* at 1:51–57, 5:4–9, 6:4–52, 6:64–7:5, 7:18–29, 27:16–47. Thus, the Specification supports a construction of a single sample solution as distinct from that single solution separated in some way, whether through aliquoting or use of beads or a solid phase.[16]

Petitioner argues that the Specification "makes clear that a 'single sample volume' is not limited to a droplet, but may be any volume, such as a 'tube or a bottle,'" and that "[n]othing in the claims or specification precludes the liquid in such volumes from subsequently being partitioned or separated—for example, using a common process known as 'droplet-based microfluidics,' where a single sample volume is analyzed by creating uniform, picoliter-to-nanoliter-sized droplets that are suspended in an immiscible fluid." Reply 10 (citing Ex. 1001, 38:63–65; Ex. 1043 ¶ 14). Petitioner argues that, indeed, the Specification states "the invention ***could*** be used with separation . . . if desired." *Id.* (citing Ex. 1001, 36:4–6).

We are not persuaded by Petitioner's argument because, although the portions of the Specification cited by Petitioner show that the claimed assay may be performed in a droplet to detect analytes in the droplet (i.e., a single sample volume may first be separated into droplets before amplification and detection is performed in a droplet), they do not suggest that the droplet, either alone or as a collection, is the same "single sample volume" as the pre-partitioned sample volume. Rather, the "single sample volume" in which the analytes are to be amplified *and* unambiguously detected is the droplet.

[16] The parties dispute whether "separation" refers only to "use of beads or a solid phase" or also includes "forming droplets." *Compare* Paper 29, 9, *with* Paper 30, 2. However, we determine that we need not resolve this dispute because our construction of the term, "wherein analytes are amplified in the single sample volume," would be the same regardless of whether separation includes separation via formation of droplets.

Petitioner argues that the prosecution history supports its claim construction position, because during prosecution "the claims were amended to remove a requirement that the assay be performed 'in a single sample volume ***without … separation***" and replace it with "broader language requiring only that amplification occur in the 'single sample volume.'" Reply 10 (citing Ex. 1029, 2).

We are not persuaded. The amendment during prosecution replaced the claim limitation that "the assay [be] capable of detecting the presence or absence of each of said at least M analytes, in any combination of presence or absence, in a single sample volume the recited detection without immobilization, separation, mass spectrometry, or melting curve analysis" with the requirements that "wherein the assay does not require the use of mass spectrometry or immobilization of said M analytes" and "wherein the analytes are amplified in the single sample volume." Ex. 1029, 2.

This amendment is consistent with the portion of the Specification Petitioner cites above, which indicates that the claimed assay could — but need not be — used with separation. *See, e.g.*, Ex. 1029, 2 (Patent Owner emphasizing in remarks accompanying the amendment that "the . . . assay and kit disclosed [in the Application] does not require 'immobilization, separation, mass spectrometry, or melting curve analysis'"). Petitioner does not persuasively explain, however, why the amendment supports a construction of "amplified in the single sample volume" as something other than amplification taking place in the same single sample volume (whether a pre-partitioned volume or an individual droplet or aliquot after partition) in which the presence or absence of each of at least M analytes are to be detected. Put another way, Petitioner does not persuasively explain why the amendment supports a construction of a collection of droplets resulting from partitioning as the same single volume as the pre-partitioned sample.

Finally, Petitioner argues that in other, related patents, Patent Owner explicitly "exclude[d] 'separation' or limit[ed] claims to 'droplets.'" Reply 10–11 (citing Paper 9, 2).

We are not persuaded. Claim 1 of U.S. Patent No. 10,068,051 B2 recites "[a] method of detecting . . . analytes . . . without requiring any step of immobilization of said polynucleotide analytes, physical separation of said polynucleotide analytes, or mass spectrometry." IPR2024-01177, Ex. 1001, 69:8–70:11. Claim 1 of U.S. Patent No. 10,770,170 B2 recites "[a] method of unambiguously detecting any unique combination of presence or absence of at least five polynucleotide analytes in a plurality of droplets," the method comprising among other steps "partitioning [a] mixture [of sample potentially comprising the analytes and hybridization probes] into said plurality of droplets." IPR2024-01178, Ex. 1001, 65:30–61.

As an initial matter, these claims in related applications do not use the phrase "amplified in the single sample volume"; thus, they do not implicate the presumption that a claim term should be construed consistently with its appearance in other places in the same claim or in other claims of the same or related patents. *Samsung Elecs. Co., Ltd. v. Elm 3DS Innovations, LLC*, 925 F.3d 1373, 1378 (Fed. Cir. 2019) (construing a term used throughout patents derived from the same parent application in the same way). Neither are we persuaded that the claims of the related patents cited by Petitioner are inconsistent with a construction of the '921 patent claims that distinguishes between a pre-partitioned sample and the droplet or collection of droplets after partition as different "single sample volumes." If anything, claim 1 of the ’170 patent appears to distinguish between the pre-partitioned volume (i.e., the mixture of sample potentially comprising the analytes and hybridization probes) and the plurality of droplets after partition by using different terms to refer to each.

Accordingly, we construe the phrase, “wherein analytes are amplified in the single sample volume,” to require that the amplification of the analytes take place in the sample volume in which the presence or absence of the at least M analytes are to be detected, without that sample volume being further partitioned prior to the amplification.

2. “F wavelength components”

Petitioner and Patent Owner disagree whether F must be at least 2. *See, e.g.*, PO Resp. 12 (“As is clear from the language of the claims and the original specification, the ’921 Patent can detect three or more analyte molecules (M≥3) using multiple colors (F≥2).”) (citing Ex. 2018 ¶ 47); Reply 7–8 (“The claim language does not require ‘F’ to be at least 2 or ‘M’ to be at least 3 as PO contends—in fact, the claim is agnostic as to the specific value of ‘F’ and ‘M’; it simply requires that ‘M is greater than F,’ meaning ‘F’ can be any value less than ‘M,’ including 1.”) (citing, *e.g.*, Pet. 23, 28; Ex. 1002 ¶¶ 111, 124; Ex. 1044, 85:9–22); Sur-reply 11.

We agree with Patent Owner that F should be construed to be at least 2. As an initial matter, and as Patent Owner points out, the claims recite that the fluorescent signal comprises “F wavelength *components*,” and “[t]here is a presumption that plural terms refer to two or more items.” Sur-reply 11 (citing *Apple Inc. v. MPH Techs. Oy*, 28 F.4th 254, 261 (Fed. Cir. 2022)). Although “[t]hat presumption can be overcome when the broader context shows a different meaning applies,” in this case neither the claim language nor the Specification suggests that the term “components” should be interpreted to also include a single component. *MPH Techs.*, 28 F.4th at 262. As in *MPH Technologies*, "[t]here is no indication, for example, that the use of the plural . . . represents an effort to 'achieve grammatical consistency' with another term" or was "used to describe a function (e.g., means for creating [wavelength components])." *Id.* (citing *In re Omeprazole Patent Lit.*, 84 F. App'x 76, 80 (Fed. Cir. 2003) (non-precedential) and *Versa Corp. v. Ag-Bag Int'l Ltd.*, 392 F.3d 1325, 1330 (Fed. Cir. 2004)). This case is also distinguishable from *Dayco*, in which a different presumption was also implicated. *Dayco Prods., Inc. v. Total Containment, Inc.*, 258 F.3d 1317, 1327–1328 (Fed. Cir. 2001) (holding that "plurality of . . . projections with recesses therebetween" does not require that there be at least three projections despite reference to "recesses" in the plural, in part because "'plurality,' when used in a claim, refers to two or more items, absent some indication to the contrary").

We are also not persuaded that the written description requires departure from the presumption that "F wavelength components" refers to two or more wavelength components (i.e., F≥2). For instance, although the Specification states that "singular forms . . . are intended to include the plural forms as well, unless the context clearly indicates otherwise," there is no language indicating the converse (i.e., that the plural should include the singular). Ex. 1001, 10:1–3; *MPH Techs.*, 28 F.4th at 262. Likewise, although Petitioner is correct that the Specification describes embodiments using a single color or wavelength (i.e., F=1), the Specification neither broadly describes "the invention" as an assay wherein the fluorescent signal comprises one or more wavelength components, nor describes the embodiment using a single wavelength component as preferred. *MPH Techs.*, 28 F.4th at 262; Reply 8–9 (citing Ex. 1001, 22:18–62, 23:38–47;

Ex. 1044, 68:15–23, 70:16–71:1). As Patent Owner correctly points out, a claim does not need to cover all embodiments. Sur-reply 13 (citing *Intamin Ltd. v. Magnetar Techs., Corp.*, 483 F.3d 1328, 1337 (Fed. Cir. 2007)).

Finally, we acknowledge Petitioner's argument that, during his deposition, Patent Owner's expert, Dr. Struhl, "unequivocally acknowledged that 'F' can be 1" in the claims. Reply 8 (citing Ex. 1044, 85:9–22). However, Dr. Struhl clarified during redirect examination that his opinion is that F is at least 2 and M is at least 3. Ex. 1044, 195:22–196:8. More importantly, even accepting Dr. Struhl's testimony that F can be 1 in the claims, we would give greater weight to the intrinsic evidence discussed above.

Accordingly, we construe F in "F wavelength components" to be at least 2.

D. Ground 1: Alleged Anticipation by Larson (claims 1–10, 13–19)

1. Overview of Larson (Ex. 1017)

Larson, titled "Digital Analyte Analysis," relates to "droplet based digital PCR and methods for analyzing a target nucleic acid using the same." Ex. 1017, codes (54), (57). Larson was filed February 11, 2011 and issued October 13, 2011. *Id.* at codes (22), (43).

Larson teaches that the ability to multiplex (i.e., "perform and quantify multiple amplifications simultaneously within the same reaction volume") has become "paramount" in PCR and real-time PCR (or qPCR) because of increasing need for "higher throughput in analyzing multiple targets in parallel in the fields of genomics and genetics" and "more efficient use of sample . . . in medically related fields such as diagnostics." Ex. 1017 ¶ 5. However, Larson teaches that "a general purpose single-pot solution to qPCR multiplexing does not yet exist." *Id.*

Larson teaches that "[d]igital PCR (dPCR) is an alternative quantitation method in which dilute samples are divided into many separate reactions." Ex. 1017 ¶ 6. Larson teaches that methods of its invention involve "novel strategies" for multiplexing "based on the singular nature of amplifications at terminal or limiting dilution that arises because most often only a single target allele is ever present in any one droplet even when multiple primers/probes targeting different alleles are present." *Id.* ¶ 45.

Larson explains that, at terminal dilution, "the vast majority of reactions contain either one or zero target DNA molecules." Ex. 1017 ¶ 6. Larson compares this situation with "limiting dilution," in which "some reactions contain zero DNA molecules, some reactions contain one molecule, and frequently some other reactions contain multiple molecules, following the Poisson distribution." *Id.*; *see also id.* ¶ 52. Larson refers to the scenario in which "multiple different target molecules co-occupy[] the same droplet" as "multiple occupancy." *Id.* ¶ 141.

Larson teaches that, using its invention, higher-plex assays may be performed (1) using a single probe color in which "each intensity level uniquely identifies a DNA target"; (2) "using multiple different probe colors" where multiple targets can also be identified based on intensity; or (3) using "multiple probes that constitute a unique signature of both colors and intensities" to identify a single target. Ex. 1017 ¶¶ 138–140.

Larson teaches that, generally, "the above three methods for higher-plex dPCR are simplest to implement under conditions of terminal dilution, . . . when the probability of multiple different target molecules co-occupying the same droplet is very low compared to the probability of any single target occupying a droplet." Ex. 1017 ¶ 141. Larson explains this is because

> [w]ith multiple occupancy arises the complexity of simultaneous assays competing within the same reaction droplet, and also complexity of assigning the resulting fluorescence intensity that involves a combination of fluorescence from two different reaction products that may or may not be equal to the sum of the two fluorescence intensities of the individual reaction products.

Id. ¶ 141. According to Larson, however, "methods of the invention can accommodate these complications arising from multiple occupancy." *Id.*

2. Analysis of Challenge to the Claims

The dispositive issue as to Ground 1 is whether Petitioner has satisfied its burden to show that Larson discloses an assay or kit "capable of unambiguously detecting the presence or absence of each analyte of M analytes in a single sample volume, in any combination of presence or absence . . . *wherein the analytes are amplified in the single sample volume*." Ex. 1001, 67:10–26 (emphasis added); *see also id.* at 68:19–38.

Petitioner asserts that paragraph 140 of Larson discloses "using two colors to identify three target analytes, relying on the combination of two colors as the multi-color label for one analyte, with each of the two colors individually as label for the other two targets." Pet. 28–29. Petitioner asserts that "Larson discloses detecting whether these three targets are present in a 'sample volume,' specifically, a portion of a sample solution that is subsequently partitioned into droplets." *Id.* at 29.

Petitioner further asserts that Larson discloses "conducting the digital PCR assay in paragraph 140 under conditions of terminal dilution, where the probability of a single target molecule in droplets is much greater than, and overwhelms[,] that of[] multiple targets." Pet. 29. Petitioner asserts that, "[u]nder these conditions, the detectable signal generated for each potential analyte is distinct and readily distinguishable, providing 'unambiguous' detection of the analytes." *Id.*

With respect to the limitation, "wherein the analytes are amplified in the single sample volume," Petitioner asserts that "Larson discloses PCR assays that amplify target polynucleotide in a single sample volume, specifically *a portion of the reaction mixture that is subsequently partitioned into droplets*." Pet. 31 (emphasis added).

There is no dispute that the amplification of the analytes in Larson occurs in the droplets. Larson ¶¶ 6–7 (explaining that "[d]igital PCR (dPCR) is an alternative quantitation method in which dilute sample are divided into many separate reactions" and that "[t]he template is amplified *in the droplet* for detection") (emphasis added). Because the claims require that amplification of the analytes be in the same sample volume in which the at least M analytes, in any combination of presence or absence, is unambiguously detected, and because the pre-partition single sample volume is not the same single sample volume as the collection of droplets after partition under our claim construction set forth above, we find Petitioner has not shown, by a preponderance of evidence, that claims 1–10 and 13–19 are anticipated by Larson.[17] Tr. 68:12–69:14 (Petitioner's counsel

[17] Petitioner asserts that, in the ChromaCode Litigation in the district court, "Patent Owner states with regard to Petitioner's products[that] 'the detection is performed in a single ***sample volume*** that is ***subsequently*** partitioned into multiple droplets." Pet. 31–32. According to Petitioner, Patent Owner alleges that the limitation "analytes are amplified in the single sample volume" is met where the sample is partitioned into individual droplets before amplification. *Id.* at 32. Petitioner asserts that, "[t]hus, Larson . . . discloses the exact thing that Patent Owner now accuses of

acknowledging that Ground 1 is dependent on a claim construction that the pre-partition single sample solution is the same single sample solution as the collection of droplets after partition).

E. Ground 2: Alleged Obviousness over Larson (claims 4, 6, 7, 14–19)

Petitioner's Ground 2 alleges that claims 4, 6, 7, and 14–19 are obvious over Larson. Petitioner's arguments with respect to this ground rely on and cross-reference its arguments in Ground 1 that Larson anticipates claims 1–10 and 13–19. Pet. 44–51. Accordingly, for the same reasons set forth in Ground 1, we find Petitioner has not shown, by a preponderance of evidence, that claims 4, 6, 7, and 14–19 are obvious over Larson.

F. Ground 3a: Alleged Obviousness over Larson (claims 1–10, 12–19)

For purposes of this ground of invalidity, Petitioner focuses on a single droplet as the "single sample volume." Pet. 54 ("Whereas in Ground 1 the 'single sample volume' is taken to be a portion of the original sample mixture prior to it being partitioned into droplets, here the 'single sample volume' is an individual droplet generated by partitioning the sample mixture").

With respect to this ground, the parties disagree as to whether Larson teaches or suggests an assay "capable of unambiguously detecting the presence or absence of . . . at least M analytes, in any combination of presence or absence, in a single sample volume . . . when M is greater than

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infringement in district court." *Id.* Even assuming this to be the case, in this proceeding we must evaluate the grounds of challenge based on the claims as properly construed, rather than based on the parties' alleged litigation positions in the district court.

F" — that is, when the number (F) of wavelengths components used to encode the analytes is greater than the number of analytes. Pet. 54–57; PO Resp. 40–54.[18]

The parties also dispute whether Petitioner has shown, by a preponderance of evidence, that an ordinarily skilled artisan would have had reason to modify Larson to arrive at the challenged claims, with a reasonable expectation of success. Pet. 57; PO Resp. 54–62.

The parties further disagree as to whether Petitioner has met its burden with respect to dependent claims 4 and 13, which require that the coding scheme generated in claim 1 to be, respectively, "made nondegenerate by enumerating every legitimate result that can be obtained from said coding scheme, identifying each legitimate result that is degenerate, and eliminating at least one potential analyte code from said coding scheme to eliminate degeneracy," or "innately nondegenerate or non-degenerate by construction." Pet. 34–35, 39–40, 44–47, 62, 64; PO Resp. 62–63.

Finally, Patent Owner argues that "several secondary considerations further demonstrate the challenged claims patentability." PO Resp. 63–67.

1. Analysis of Claim 1

For the reasons discussed below, we are not persuaded Petitioner has shown, by a preponderance of evidence, that claim 1 is unpatentable as obvious over Larson.

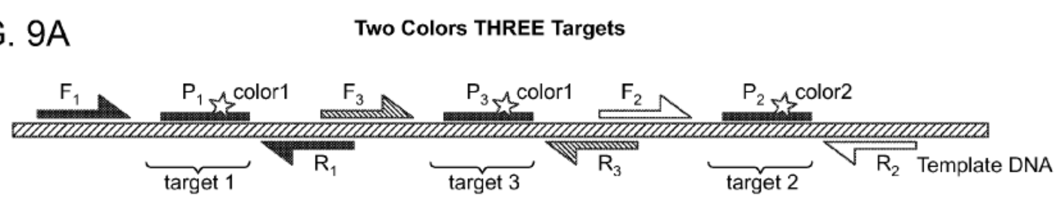
[18] Because we have construed F to be at last 2, the number of analytes (M) must be at least 3 in the claims.

a. *"capable of unambiguously detecting the presence or absence of each of . . . at least M analytes, in any combination of presence or absence, in a single sample volume . . . when M is greater than F"*

Petitioner asserts Larson teaches scenarios where individual droplets have multiple target molecules, because "the number of targets present in droplets will follow a Poisson distribution" in situations of limiting dilution. Pet. 54–55 (citing Ex. 1017 ¶ 6; Ex. 1002 ¶ 331). Petitioner asserts that Larson's Figure 9 describes an assay that allows detection of analytes in such "multiple occupancy" droplets using fewer fluorescent wavelength components than the number of analytes in the droplet (i.e., where M is greater than F), because Figure 9 teaches using two different intensities of one color to identify two targets and a second color to identify the third target. Pet. 53–56 (citing, *e.g.*, Ex. 1017 ¶ 121, Figs. 9A, 9C; Ex. 1002 ¶¶ 324–327, 333–335).

For ease of reference, Larson's Figure 9A is reproduced below:

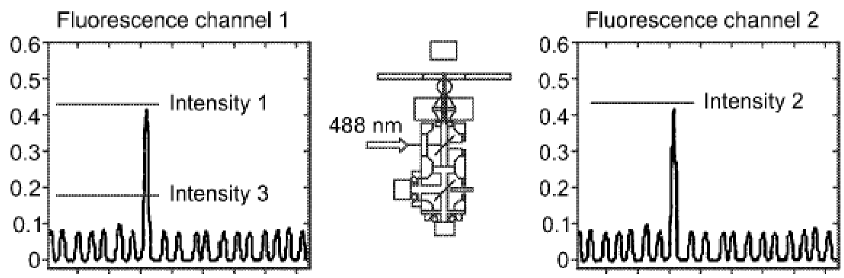
FIG. 9A



Ex. 1017, Fig. 9A. Figure 9A shows a template DNA amplified with three sets of forward (F1, F2, and F3) and reverse (R1, R2, and R3) primers, where probes (P1, P2 and P3) labeled with fluorophores (color 1, color 2, and color 1, respectively) bind to target genetic sequences target 1, target 2, and target 3. *Id.* ¶ 121.

Larson's Figure 9C is reproduced below:

FIG. 9C



Id., Fig. 9C. Figure 9C shows that, after microdroplets are made of diluted solution of template DNA under conditions of limiting or terminal dilution, "[m]icrodroplets containing target sequence 1 or 3 emit fluorescence of color 1 at two different intensities' and microdroplets containing target sequence 2 emit fluorescence of color 2." *Id.* ¶ 121.

Petitioner provides the following table as a representation of the fluorescence "signal signatures of the three individual target in Figure 9" based on Figure 9C:

	Color 1	**Color 2**
Intensity slightly less than 0.2	Target 3	NA
Intensity slightly greater than 0.4	Target 1	Target 2

Pet. 56 (citing Ex. 1002 ¶ 335).

Petitioner argues that, because each target in Figure 9 has its own unique color/intensity signature that is not overlapping with that of the other two targets, and because the intensity of each of the targets is additive, any combination of the presence or absence of the targets would "each sum to unique values in each of the two colors used." Pet. 56–57 (citing Ex. 1002 ¶¶ 335–337, Fig. 9C). Petitioner asserts that, accordingly, the encoding scheme disclosed in Figure 9 "allows for determination of precisely which targets are present in a droplet even when multiple targets are present in a droplet." *Id.* at 56 (citing Ex. 1002 ¶¶ 335–337). Petitioner further asserts that this encoding scheme supports Larson's teaching that its invention is "compatible with all distributions of DNA loading that conform to limiting or terminal dilution," including accommodating complications arising from multiple occupancy. *Id.* at 52, 55–57 (citing, *e.g.*, Ex. 1017 ¶¶ 52, 121, 129, 141, Fig. 9A; Ex. 1002 ¶¶ 321, 332, 333, 335–337).

Although we agree with Petitioner that Larson contemplates scenarios in which droplets may contain more than one target molecule and does not require the exclusion of such multiple occupancy droplets in all embodiments, we are nevertheless not persuaded Petitioner has shown, based on a preponderance of evidence, that Larson renders obvious an assay capable of unambiguously detecting, in such multiple occupancy droplets, the presence or absence of each of at least M analytes, in any combination of presence or absence, when M is greater than F and F is at least 2.

Petitioner's obviousness challenge is based on Larson allegedly rendering obvious unambiguous detection of multiple *molecules* in a single droplet, rather than unambiguous detection of multiple *targets* on a single molecule. Tr. 17:25–18:12 (Petitioner's counsel asserting that "what [Petitioner has] always thought about for the purposes of this position is to analyze multiple DNA molecules," rather than trying to "rely upon . . . multiple sequences on the same DNA molecule"); Ex. 2025, 30:8–11 (Petitioner's expert agreeing to a distinction between "a single analyte molecule" and "multiple targets within that single molecule").

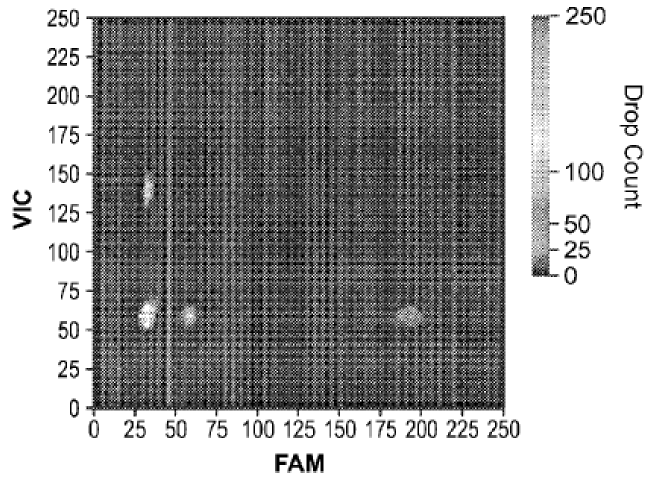
As Patent Owner points out, however, Larson does not suggest the coding scheme in Figure 9 as a method of accommodating complications arising from multiple occupancy (i.e., multiple molecules in a droplet); rather, it "illustrates a ***single template DNA molecule*** which includes three targets." PO Resp. 49; *see also* Ex. 1017 ¶ 121 (explaining that, "in Panel A of FIG. **9**, *a* template DNA is amplified with three sets of primers") (emphasis added), Fig. 9B (explaining that "*single* molecule amplification produces . . . fluorescence burst of a quantized intensity") (emphasis added); Ex. 2017, 210:16–25 (explaining that Figure 9A shows multiple targets on a single molecule, which would "segregate into a single droplet"). Stated somewhat differently, Larson's Figure 9 relates to a sample volume in which multiple targets may be detected on a ***single*** molecule, not a multiple occupancy sample having at least three target molecules as specified in claim 1.

We are not persuaded Petitioner has sufficiently explained how a method of multiplex detection of three targets on a single DNA template molecule in dPCR, as described in Figure 9, would suggest to an ordinarily skilled artisan a method for unambiguous detection of any combination of presence or absence of three target molecules in a multiple occupancy droplet, particularly when, as Petitioner agrees, Larson "focuses" on scenarios where droplets have only one target molecule. Pet. 54.

Indeed, as Patent Owner points out, Figure 9 does not appear to contemplate detecting droplets with more than one target sequence. PO Resp. 53–54. Instead, Larson explains that "[m]icrodroplets containing target sequence 1 *or* 3 emit fluorescence of color 1 at two different intensities; and microdroplets containing target sequence 2 emit fluorescence of color 2." Ex. 1017 ¶ 121 (emphasis added). Similarly, Figure 9D, reproduced below, is a histogram showing "[t]he number of microdroplets containing target 1, 2 *or* 3," *id.* (emphasis added), without identification of any microdroplets containing more than one target:

FIG. 9D

Histogram showing four populations of droplets; with target1, with target2, with target3 and without any of the three targets.



Id., Fig. 9D.

In this regard, we acknowledge Dr. Batt's testimony that Figure 9D only "plot[s] the fluorescence intensity from two different fluorophores" and does not "necessarily portray every single droplet." Ex. 2017, 111:21–116:14. However, to the extent Figure 9D does not exclude the possibility that there are, in fact, droplets containing target 1, target 2, and target 3 in each possible combination of presence or absence, it also does not suggest that Larson teaches or suggests an assay capable of unambiguously *detecting* populations of such droplets. If anything, it suggests the contrary.

We further acknowledge Petitioner's response in the Reply that Larson describes "methods *adaptable* for 'multiple occupancy'" and that "Larson . . . notes that '[i]n the simplest case targets are unlikely to reside on the same DNA fragments, such as when targets are from different cells,' and discloses primer/probe pairings for such cases." Reply 1 (emphasis added), 14. We also give some weight to Dr. Batt's testimony that, even though Figure 9C "in and of itself doesn't show . . . multiple occupancy," "the technique displayed . . . clearly indicates how you could account for multiple occupancy." Ex. 2017, 209:25–212:11 (testifying that "the approach [of Figure 9] definitely does show multiple occupancy" because "if you had both target 1 and target 3, . . . you would see that as an intensity of somewhere around .6" and if "you had target 1 and target 2, you would . . . have a combination of these two fluorescences shown in the left-hand panel and right-hand panel [of Figure 9C] at the same droplet"). Finally, we recognize that the level of skill in the art is high. *See, e.g.*, *supra* Section II.B.; Tr. 72:24–73:3 (arguing that multiplex PCR was a "standard technique in the toolkit of the skilled artisan" and that such an artisan "knew how to overcome [the] challenges" of doing multiple PCRs in the same reaction identified in Larson).

However, we balance the consideration of the above evidence and arguments with the teachings of Larson that explicitly recognize the distinct issues posed by analyte detection in multiple occupancy molecules.[19] For instance, Larson suggests, as one of the advantages of dPCR, "the singular nature of amplifications at terminal or limiting dilutions that arises because most often only a single target allele is ever present in any one droplet even when multiple primers/probes targeting different alleles are present." Larson ¶ 45. Larson teaches "[t]his alleviates the complications that otherwise plague simultaneous competing reactions, such as varying arrival

[19] We note that, although the parties' agreed definition of POSA stated that such a person would have understood "[t]echniques for labeling probes and detecting multiple targets using fewer differently colored fluorophores than polynucleotide targets," this understanding was not specifically related to detection of at least three analyte molecules in a dPCR droplet formed according to Larson's method. Reply Br. 11; *supra* Section II.B.

time into the exponential stage and unintended interactions between primers." *Id.* Similarly, Larson teaches that "[w]ith multiple occupancy arises the complexity of simultaneous assays competing within the same reaction droplet, and also complexity of assigning the resulting fluorescence intensity that involves a combination of fluorescence from two different reaction products that *may or may not be* equal to the sum of two fluorescence intensities of the individual reaction products." *Id.* ¶ 141 (emphasis added).

In the Reply, Petitioner also points to Larson's disclosure of "a method in which a single droplet contains four probes [of two colors] to detect three targets (T1, T2, T3)," and further argues Larson teaches that the "intensity of signals from color labels can be adjusted to prevent ambiguous results, specifically to avoid overlap between signals from droplets containing different target analytes." Reply Br. 14 (citing Ex. 1017 ¶¶ 35, 140–142, 146, 150; Ex. 1002 ¶ 163). Petitioner asserts that "[t]hese methods directly address the challenge of multiple occupancy and enable clear unambiguous detection of all analyte combinations within a single droplet." *Id.*

We are not persuaded. Larson itself does not describe the teachings above as "directly" addressing challenges of multiple occupancy. For instance, although Larson teaches "[h]igher-[p]lex [r]eactions" in which a droplet may contain four probes to detect three targets, Larson specifically teaches that these methods are "simplest to implement under conditions of terminal dilution, that is when the probability of multiple different target molecules co-occupying the same droplet is very low compared to the probability of any single target occupying a droplet." Ex. 1017 ¶¶ 138–141.

As noted above, moreover, Larson attributes this to the complications that may arise in connection with multiple occupancy. *Id.* ¶ 141.

Likewise, although Larson teaches methods to optimize resolutions of populations of droplets containing different target molecules based on fluorescent color and intensity in a two-dimensional histogram, the detected populations are of droplets containing single targets (e.g., TERT, E1a, RNaseP, 815A, 5C, 88G, 815G, 5G, and 88A, respectively). Petitioner fails to persuasively explain why it would have been obvious to translate the ability to perform such optimization in single-target droplet populations to also include populations of multiple occupancy droplets, which implicate complications beyond resolutions of fluorescence signals from a single reaction product. Ex. 1017 ¶¶ 45 (discussing complications arising from simultaneous competing reactions), 141 (discussing complications relating to, e.g., competition of reactions within the same droplet and assignment of fluorescence intensity resulting from combination of fluorescence from two different reaction products).

In this regard, we acknowledge Larson's teachings that the methods of its invention "can accommodate . . . complications arising from multiple occupancy" and are "compatible with all distributions of DNA loading that conform to limiting or terminal dilution." Reply 13 (citing Ex. 1017 ¶¶ 52, 141). We are not persuaded, however, that these general statements, which are silent as to *how* multiple occupancy complications are "accommodated," would have suggested to an ordinarily skilled artisan that Larson's multiplex coding schemes for detecting multiple targets in a dPCR sample — which most often contain a single molecule — may be used to unambiguously detect three or more analytes, in any combination of presence or absence, in multi-occupancy droplets. For instance, Larson appears to suggest that, while in "a simple duplex system" the contribution from each of the two probes may be recovered, "higher degrees of multiplexing would require greater dilution." Ex. 1017 ¶ 129. Thus, the accommodation of complications arising from multiple occupancy may simply be one of greater dilution — a rational interpretation, given Larson's focus on methods for assaying single–occupancy samples. In other words, Larson's brief reference to "multi-occupancy droplets," absent instructions beyond "greater dilution," is not shown on this record to suggest a different method for accounting for the complexities arising in samples that include three or more molecules. *Id*.

In the Reply, Petitioner also relies on Larson's Figure 11 and its accompanying explanation to show that Larson discloses an assay capable of detecting single droplets containing more than one target. Reply Br. 15–16. Figure 11a is reproduced below for ease of reference:

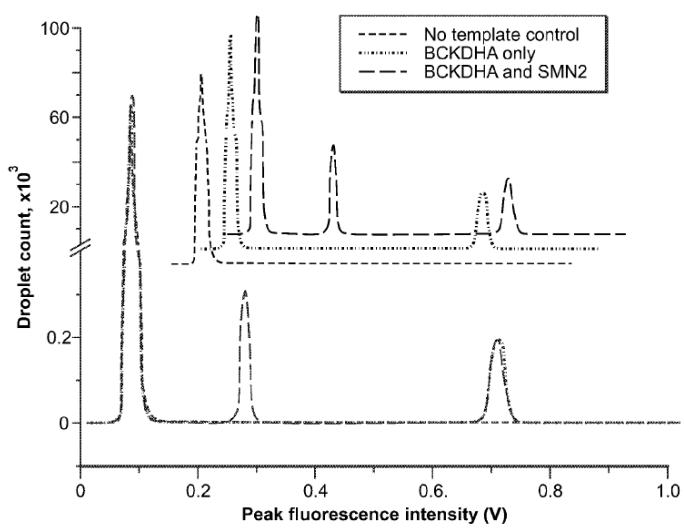


FIG. 11a

Ex 1017, Fig. 11a. Figure 11 is a histogram showing the number of droplets having particular peak intensities where a single fluorophore (FAM) was used to detect both a reference (BCKDHA) and a target (SMN2) DNA in a gene copy assay. *Id.* ¶ 130.

With respect to Figure 11, Larson states that, in the case where the sample contained the target and reference genes in a 1:1 ratio, peaks at 0.08, 0.27, and 0.71 V were readily apparent, representing droplets containing, respectively, no DNA, the target SMN2 gene, and the reference BCKDHA gene. Ex. 1017 ¶ 131. Larson further states, however, that "[a] very small peak appeared at ~0.9V, not visible on the scale of FIG. 11a, that corresponded to droplets occupied by both genes." *Id.*

We acknowledge that, unlike the other examples in Larson cited by Petitioner, Larson describes the assay used for Figure 11 as capable of detecting multiple occupancy molecules (i.e., droplets occupied by both SMN2 and BCKDHA genes). We note, however, that Figure 11 and its accompanying text describes a droplet containing two molecules (i.e., a droplet with double occupancy), whereas under our construction of claim 1, the number of analytes must be at least 3. Even if we included Figure 11 and the accompanying text in our analysis,[20] and even assuming Figure 11 discloses an assay capable of using a single fluorophore to unambiguously detect the presence or absence of two analytes in any combination, Petitioner has not persuasively explained why this would have rendered obvious an assay capable of unambiguously detecting the presence or absence of *three* analytes in any combination using fewer than three colors.[21] For instance,

[20] Patent Owner has moved to strike parts of the Reply discussing among other things, "portions of Larson not previously argued, including Figure 11a and ¶¶130-131." Mot. 1. We address the Motion to Strike below.

[21] In a duplex reaction involving two analytes A and B, any combination of the presence or absence of the analytes would involve only four populations

Larson itself distinguishes between "a simple duplex system" involving two analytes and higher degree of multiplexing, suggesting that "[h]igher degrees of multiplexing would require greater dilution." Ex. 1017 ¶ 129.

In short, Petitioner's obviousness arguments in Ground 3a require more than a showing that Larson's methods would result in multiple occupancy droplets and that Larson discloses an encoding scheme in which any combination of the presence or absence of at least three targets would "each sum to unique values in each of the two colors used." Pet. 56–57 (citing Ex. 1002 ¶¶ 335–337, Fig. 9C). Rather, they require a showing of obviousness as to a method of *unambiguously detecting* the presence or absence of *three* or more analytes, in *any* combination of presence or absence, in a *droplet* produced accordingly to Larson's dPCR method. We are not persuaded the Petitioner has made the requisite showing with respect to this limitation.

b. Reason to modify and reasonable expectation of success

We further agree with Patent Owner that Petitioner has not shown, by a preponderance of evidence, that an ordinarily skilled artisan would have had reason to combine Larson's disclosures to arrive at the claimed assay, with a reasonable expectation of success.

Petitioner asserts that an ordinarily skilled artisan "would have been highly motivated to identify unambiguously which targets (or combinations thereof) are present in a droplet so that one extracts the maximum amount of

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of droplets: no analyte (i.e., neither A nor B); A; B; or A and B. In a reaction involving three analytes, A, B, and C, however, any combination of the presence or absence of the analytes would involve eight droplet populations (no analyte; A; B; C; A and B; A and C; B and C; and A, B, and C).

information from the experiment and adequately deals with the 'multiple occupancy' situation that arise with loading of targets into droplets with a Poisson distribution," and would have a reasonable expectation of success of doing so in light of Larson's teachings. Pet. 57 (citing Ex. 1002 ¶¶ 331–338).

However, Petitioner does not adequately explain what additional information an ordinarily skilled artisan would have expected, or been motivated to obtain, from the multiple occupancy droplets that could not or would not have been obtained from analysis of the collection of droplets as a whole, as taught in Larson.

As Patent Owner argues, and as Petitioner acknowledges in its Ground 1 anticipation challenge, Larson's method is able to detect the analytes present in a sample by dividing the sample into droplets and separately analyzing the single-occupancy droplets, without relying on additional analysis of multiple occupancy droplets. PO Resp. 56–58 (arguing that, given the rarity of multiple occupancy droplets at limiting or terminal dilution, Petitioner's expert "fails to explain why the POSA would have thought that additionally analyzing the . . . multiple occupancy droplets would have any meaningful 'impact' on the 'total data collected in that experiment'" or on cost savings, particularly when "a POSA readily would have recognized that . . . the single-analyte droplets provide all the information required to identify the analytes present in the [original] sample") (citing, *e.g.*, Ex. 2017, 175:16–177:14, 179:14–180:7; Ex. 2018 ¶¶ 124–125). The lack of persuasive evidence as to what benefit would have been obtained by additionally analyzing multiple occupancy droplets in Larson's method is particularly problematic given Larson's teaching of the potential complications arising from multiplex analyte detection in multiple occupancy droplets. *See, e.g.,* PO Resp. 60–61; *Medichem, S.A. v. Rolabo, S.L.*, 437 F.3d 1157, 1165 (Fed. Cir. 2006) (explaining that "the benefits, both lost and gained, should be weighed against one another" in determining whether there is a basis to modify the disclosure of a prior art reference).

In this regard, we note that this is not a case where "a motivation to combine the steps was not required because the . . . steps were disclosed in a single embodiment in a single prior art reference." *Guardant Health, Inc. v. Univ. of Washington*, 2026 WL 184334, *4 (Fed. Cir. 2026). As noted above, we are not persuaded that Larson discloses an embodiment of an assay capable of "unambiguously detecting the presence or absence" of each of at least 3 analytes, in any combination of presence or absence, in a droplet containing at least three molecules.[22]

We are also not persuaded that Petitioner has shown, by a preponderance of evidence, that an ordinarily skilled artisan would have had

[22] To the extent Petitioner argues inherency, we are not persuaded. Tr. 27:3–20 (Petitioner counsel states that "[a] person of ordinary skill in the art [would be] motivated to use Figure 9 with multiple targets. I mean, it's unavoidable. [W]hen . . . you're doing Poisson loading, you can't avoid it, it's going to be there."). As an initial matter, Ground 3a in the Petition was not based on an inherency argument, and Petitioner must "adhere to the requirement that the initial petition identify 'with particularity' the 'evidence that supports the grounds for the challenge to each claim.'" *Intelligent Bio-Sys.*, 821 F.3d at 1369 (quoting 35 U.S.C. § 312(a)(3)). Furthermore, inherency "must inevitably result from the disclosed steps" and "may not be established by probabilities or possibilities." *In re Montgomery*, 677 F.3d 1375, 1379–80 (Fed. Cir. 2012) (internal quotation marks omitted). Assuming that Larson's method would inherently result in multiple occupancy droplets comprising at least 3 analytes, Petitioner has not persuasively explained how using the encoding scheme in Figure 9 would inevitably result in an assay capable of unambiguously *detecting* the at least 3 analytes, in any combination of presence or absence, in a droplet produced according to Larson's method.

a reasonable expectation of success of combining the disclosures of Larson to arrive at the claimed assay. Petitioner relies on paragraphs 331–338 of Dr. Batt's declaration to support its arguments with respect to motivation to combine and reasonable expectation of success. Pet. 57 (citing Ex. 1002 ¶¶ 331–338); Tr. 70:19–73:3. Dr. Batt's testimony regarding reasonable expectation of success, however, relies on his opinion that Larson itself discloses to a POSA, in the form of Figure 9, "an assay that determines without ambiguity the presence or absence of three different targets (and combinations thereof) in individual droplets." Ex. 1002 ¶ 337. For the reasons discussed above in Section II.F.1.a, however, we do not find that Figure 9, which illustrates a sample containing three targets on one molecule, discloses an assay capable of detecting targets in samples containing at least three molecules, as recited by the claims.

Finally, we acknowledge Petitioner's arguments that Dr. Struhl's opinions should be accorded no weight. Reply 2–6. Having reviewed the record as whole, we decline to entirely disregard Dr. Struhl's testimony, particularly where we find the testimony to be corroborated by other evidence, including Larson itself. To the extent Petitioner alleges any discrepancy between Dr. Struhl's deposition testimony and his declaration, we have taken such alleged discrepancies into account and accorded the opinions appropriate weight.

2. Analysis of claim 14

Claim 14 recites "[a] kit for detecting the presence or absence of at least M analytes by generating a single cumulative measurement, comprising analyte-specific reagents." Ex. 1001, 68:19–38. Claim 14 differs from claim 1 in that it does not recite limitations requiring that fluorescent signal encoding the M analytes comprise F wavelength components where M is greater than F. *Compare id.*, *with id.* at 67:9–25.[23] However, in its analysis, Petitioner does not distinguish claim 14 from claim 1 based on these differences; instead, in discussing claim 14 the Petition generally cross-references its analysis of claim 1. Pet. 64–65 (cross-referencing Pet. Sections XI.L, X.II.D., X.III.A.4–6; Ex. 1002 ¶¶ 364–385). Accordingly, we determine that, for the same reasons discussed above with respect to claim 1, Petitioner has not shown, by a preponderance of evidence, that claim 14 is obvious over Larson.

3. Dependent claims

Because we determine Petitioner has not shown, by a preponderance of evidence, that claims 1 and 14 are obvious over Larson, we likewise find Petitioner has failed to carry its burden with respect to the obviousness of dependent claims 2–10, 12, 13, and 15–19 over Larson. *See, e.g.*, *In re Fritch*, 972 F.2d 1260, 1266 (Fed. Cir. 1992) ("[D]ependent claims are nonobvious if the independent claims from which they depend are nonobvious.").

4. Objective indicia of non-obviousness

Because we determine Petitioner has not shown, by a preponderance of evidence, that claims 1–10 and 12–19 are obvious over Larson, we need not, and do not, address Patent Owner's arguments regarding objective indicia of non-obviousness.

[23] Claim 15 depends from claim 14 and recites these additional limitations (i.e., "wherein the fluorescent signal comprises F wavelength components and said kit is capable of unambiguously detecting the presence of absence of M analytes, in any combination of presence or absence, when M is greater than F"). Ex. 1001, 68:39–43.

5. Conclusion

For the reasons discussed above, we determine Petitioner has not shown, by a preponderance of evidence, that claims 1–10 and 12–19 are obvious over Larson.

G. Ground 3b: Alleged Obviousness over Larson and Saxonov (Claims 1–10, 12–19)

Petitioner argues that, "[t]o the extent Patent Owner contends that Larson alone does not render obvious the unambiguous detection of multiple targets in a single droplet, it would have been obvious in further view of Saxonov," which Petitioner asserts similarly teaches loading droplets with targets according to a Poisson distribution and, thus, contemplates the possibility of multiple targets in a droplet. Pet. 58 (citing Ex. 1004 ¶¶ 3, 43; Ex. 1002 ¶¶ 339–348), 64–65 (referring back to analysis of claim 1 to argue that claim 14 is also, in the alternative, rendered obvious by Larson in further view of Saxonov).[24]

Saxonov, titled "Digital Assays with Multiplexed Detection of Two or More Targets in the Same Optical Channel," was filed July 12, 2012, and claims the benefit of provisional application Nos. 61/507,082, filed July 12, 2011 and 61/510,013 (Ex. 1007, "the '013 provisional"), filed July 20, 2011. Ex. 1004, codes (22), (54), (60). To assert Saxonov as prior art to the '921 patent, Petitioner relies on the filing date of the '013 provisional. Pet. 13 (citing Ex. 1004, code (60)).

[24] Petitioner also relies on the combination of Larson and Saxonov, as an alternative to Larson alone, to render obvious the additional limitation of dependent claim 12, "wherein at least one step of said assay is performed using a computer and wherein said computer is connected to a thermal cycler." Pet. 62–64.

The parties dispute whether Saxonov is available as prior art under pre-AIA 35 U.S.C. § 102(e). A reference patent or published application may be entitled to the benefit of its provisional application's filing date for pre-AIA 35 U.S.C. § 102(e) prior art purposes when two conditions are met: (1) the provisional provides sufficient support for at least one claim in the reference patent or published patent application; and (2) the provisional application provides sufficient support for the subject matter relied upon for prior art purposes in the reference patent or application. *See Dynamic Drinkware, LLC v. Nat'l Graphics, Inc.*, 800 F.3d 1375, 1377, 1381–82 (Fed. Cir. 2015). The burden of production is on the Petitioner to show in its Petition that the asserted reference qualifies as prior art. *See id.* at 1380.

In the Petition, Petitioner principally relies on Saxonov's Figure 10 and the accompanying text in paragraphs 18, 78, and 79. Pet. 58–59 (citing Ex. 1004, Fig. 10, ¶¶ 18, 78–79; Ex. 1002 ¶ 341). In the Institution Decision, we preliminarily determined that Saxonov Figure 10 and paragraphs 18, 78, and 79 do not qualify as prior art to the '921 patent, because "Petitioner has not shown sufficiently that the Petition explains how each of [these] disclosures, . . . which are not present in the '013 provisional, are supported by the '013 provisional." Inst. Dec. 29.

In the Reply, Petitioner asserts that Patent Owner "relies exclusively on the Board's preliminary determination that certain portions of Saxonov are not prior art—a position that . . . is incorrect." Reply 12, 18–19. Petitioner argues that "the vast majority" of Saxonov's other disclosures, including Figure 1–8A and 8B, and paragraphs 2–16, 19–36, 38–53, 45–68, and 72–75, are substantially identical to those in the '013 provisional, and that these "'carried over' portions" of the '013 provisional, in combination with Larson, suffice to render claims 1–10 and 12–19 obvious. Reply 19–22 (citing, *e.g.*, Ex. 1007, 9:15, 5:19–6:3, 10:8–11:13, 15:12–17:3; Ex. 1044, 167:1–170:9; Ex. 2001 ¶¶ 88, 89; Ex. 2007). Critically, however, Petitioner does not explain in the Reply how the ’013 provisional supports Saxonov Figure 10 and paragraphs 18, 78, and 79, much less where such explanation may be found in the Petition.

Accordingly, for the reasons discussed in our Institution Decision, we determine that Petitioner has failed to meet its burden to show Saxonov Figure 10 and paragraphs 18, 78, and 79 qualify as prior art to the ’921 patent. Inst. Dec. 26–29. Furthermore, because the Petition principally relies on the above non-prior art portions of Saxonov in arguing that Larson further in view of Saxonov renders claims 1–10 and 12–19 obvious, we find that Petitioner also has not shown, by a preponderance of evidence, that claims 1–10 and 12–19 are obvious in view of Larson and Saxonov.

To the extent Petitioner argues that the portions of Saxonov directly “carried over” from the ’013 provisional suffice to render claims 1–10 and 12–19 obvious in combination of Larson, this argument is unavailing.

As noted above, the Petition principally rely on Figure 10 and paragraphs 18, 78, and 79 in arguing that claims 1–10 and 12–19 are obvious over Larson and Saxonov. Pet. 60 (“To the extent Patent Owner contends that Larson alone does not contemplate unambiguously distinguishing among droplets with different combinations of droplets *[sic]*, it would have been obvious to use the approach in Figure 10 of Saxonov with Larson.”). Excising Figure 10 and paragraphs 18, 78, and 79 from the Petition’s discussion of Saxonov, and substituting them with other portions at most peripherally discussed in the Petition, if at all, substantially alters the thrust of the Petitioner’s ground of challenge. Because “[i]t is of the utmost importance that petitioners in the IPR proceedings adhere to the requirement that the initial petition identify 'with particularity' the 'evidence that supports the grounds for the challenge to each claim,'" we decline to engage in such analysis. *Intelligent Bio-Sys., Inc.*, 821 F.3d at 1369 (quoting 35 U.S.C. § 312(a)(3)).

We are similarly unpersuaded by Petitioner's argument that Patent Owner's "response on obviousness of Saxonov and Larson has been waived." Tr. 28:5–18. Patent Owner states at various points in its Response that Petitioner has failed to establish that Saxonov is prior art, in light of the Board's preliminary determination in this regard. PO Resp. 2, 15–16, 27, 40. We find that these statements sufficiently preserve Patent Owner's argument that Petitioner fails to carry its burden of showing Saxonov to be prior art in the Petition.

H. Ground 4: Alleged Obviousness over Larson, Saxonov, and Silverbrook (claim 11)

Claim 11 depends from claim 1 and further comprises "transmitting information concerning the presence or absence of at least one of said analytes through a computer network." Ex. 1001, 68:10–12. Petitioner relies on its arguments that "Larson anticipates and/or renders obvious alone, or in combination with Saxonov, . . . the assay of claim 1." Pet. 66. Petitioner further asserts that "[c]laim 11 would have been obvious in further view of Silverbrook's teaching of a nucleic acid testing device and method that employs PCR and hybridization to identify which analytes are present in a sample and transmitting results of the testing through a computer network." *Id.* (citing, *e.g.*, Ex. 1002 ¶ 390; Ex. 1009 ¶¶ 195, 198, 201, Fig. 99).

Because we find that Petitioner has not shown, by a preponderance of evidence, that claim 1 is anticipated by Larson or rendered obvious over

Larson alone or in combination with Saxonov, and because Petitioner does not rely on Silverbrook to address these deficiencies of Larson and Saxonov as to claim 1, we find Petitioner also has not shown, by a preponderance of evidence, that claim 11 is obvious over Larson, Saxonov, and Silverbrook.

III. PATENT OWNER'S MOTION TO STRIKE

In its Motion to Strike, Patent Owner seeks to strike the following portions of Petitioner's Reply that Patent Owner asserts "improperly introduce new arguments and evidence based on":

> (1) portions of Larson not previously argued, including Figure 11a and ¶¶130-31 (Paper 20 at 1-2, 15-16, 22);
>
> (2) portions of Saxonov not previously argued, including FIG. 4 and ¶¶20 and 41 and corresponding matter in the Saxonov Provisional, 5:19–6:3, 9:14–15, 10:18-11:13, 15:12–17:3, and FIG. 4 (Paper 20 at 1, 18-22); and
>
> (3) Dr. Struhl's testimony relating to the above-noted sections of Larson, Saxonov, and the Saxonov provisional (Paper 20 at 1-2, 15-16, 22).

Mot. 1.

Petitioner argues that the Motion should be denied because (1) the Reply merely "elaborates on the same invalidity theories and evidence set forth in the Petition"; (2) Petitioner's discussion of Larson, Saxonov, and Dr. Struhl's testimony at issue in the Motion "is a direct and proper response to the mischaracterization of those references advanced in Patent Owner's Response"; and (3) "there is no dispute that Patent Owner has had notice and opportunity . . . to respond to any such evidence." Opp. 1.

As noted above, we determine that, even if we were to consider Larson's Figure 11 and accompanying text, including Petitioner's arguments in the Reply and Dr. Batt's and Dr. Struhl's testimony thereon, we still find that Petitioner has not shown, by a preponderance of evidence, that claims 1–10 and 12–19 are obvious over Larson. *See supra* Section II.F. We, therefore, do not decide whether discussion of Larson's Figure 11 is properly within the scope of the Reply. In order to preserve the record for appeal, however, we deny Petitioner's Motion to Strike with respect to the portions of the Reply relating to Larson's Figure 11.

Similarly, we do not rely in our analysis on Figure 4, paragraphs 20 and 41 of Saxonov, or Petitioner's arguments in the Reply and Dr. Batt's and Dr. Struhl's testimony based thereon. Accordingly, we dismiss as moot Patent Owner's Motion to Strike with respect to the portions of the Reply relating to Saxonov's Figure 4 and paragraphs 20 and 41.

IV. PETITIONER'S OBJECTIONS TO DEMONSTRATIVES

Petitioner objected to Patent Owner's demonstratives 37–39 as prejudicial and as "reflecting untimely argument and evidence." Paper 37. As the parties' demonstratives are not evidence, we do not cite them or otherwise rely on them in this Decision. Accordingly, we dismiss Petitioner's objections to Patent Owner's demonstratives as moot.

V. CONCLUSION

Based on the information presented, we conclude that Petitioner has not demonstrated by a preponderance of evidence that claims 1–19 are unpatentable. We deny in part and dismiss as moot in part Patent Owner's Motion to Strike. We further dismiss Petitioner's Objections to Demonstratives as moot.

In summary:

Claims	35 U.S.C. §	Reference(s)/ Basis	Claims Shown Unpatentable	Claims Not Shown Unpatentable
1–10, 13–19	102	Larson		1–10, 13–19
4, 6, 7, 14–19	103	Larson		4, 6, 7, 14–19
1–10, 12–19	103	Larson		1–10, 12–19
1–10, 12–29	103	Larson, Saxonov		1–10, 12–29
11	103	Larson, Saxonov, Silverbrook		11
Overall Outcome				1–19

VI. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that Petitioner has not demonstrated by a preponderance of the evidence that claims 1–19 of U.S. Patent No. 11,827,921 B2 are unpatentable; and

FURTHER ORDERED that Patent Owner's Motion to Strike is dismissed as moot in part and denied in part; and

FURTHER ORDERED that Petitioner's Objections to Demonstratives are dismissed as moot; and

FURTHER ORDERED that, because this is a final written decision, parties to this proceeding seeking judicial review of our Decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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